

Bankruptcy Prediction Pipeline

Polish Companies Dataset | XGBoost + Optuna | Documentation

1. Project Overview

This project builds a machine learning pipeline to predict corporate bankruptcy using financial ratios from Polish companies. The dataset spans 2000–2013 and contains five forecasting horizons (1–5 years). The final model targets a **2-year prediction horizon** (3year dataset) and is designed as a **first-stage screening filter** for bank credit risk assessment.

Dataset Summary

Dataset	Instances	Bankrupted	Healthy	Horizon
1year	7,027	271	6,756	5 years
2year	10,173	400	9,773	4 years
3year	10,503	495	10,008	3 years
4year	9,792	515	9,277	2 years
5year	5,910	410	5,500	1 year

2. Missing Value Treatment

The dataset contains significant missing values in two features: **Attr37** (39% missing) and **Attr21** (23% missing). Other features have less than 5% missing values.

MICE Imputation Strategy Comparison

Strategy	Test R ² (corrected)	Decision
Median imputation	-0.096	Rejected
MICE + BayesianRidge	-51.94	Rejected
MICE + RandomForest	0.871	Selected

Note: R² was corrected after fixing an index alignment bug in the validation experiment. After fix (preserving original DataFrame index), correct R² = 0.871.

Winsorization was applied before imputation to clip extreme outliers (e.g. Attr37 max = 398,920 vs median ~1.5). Imputer is fitted only on training data to prevent data leakage.

3. Hyperparameter Optimization (Optuna)

Optuna optimized XGBoost hyperparameters separately for each yearly dataset over 50 trials.
Objective metric: **F2 score** (recall weighted 2× over precision).

Parameter	Search Range	Role
n_estimators	100 – 1000	Number of trees
max_depth	3 – 10	Tree depth
learning_rate	0.01 – 0.3 (log)	Step size shrinkage
subsample	0.5 – 1.0	Row sampling per tree
colsample_bytree	0.5 – 1.0	Feature sampling per tree
min_child_weight	1 – 10	Minimum leaf sample weight
gamma	0 – 5	Minimum split loss
scale_pos_weight	1x – 3x class ratio	Class imbalance correction

4. Model Results

Performance Across All Years (threshold = 0.1)

Year	ROC-AUC	Precision	Recall	F2	Horizon
1year	0.8703	0.0753	0.9815	0.2880	5 years
2year	0.8535	0.0736	0.9500	0.2811	4 years
3year	0.8619	0.1009	0.9091	0.3494	2 years
4year	0.8269	0.1149	0.8738	0.3766	2 years
5year	0.9075	0.1734	0.9390	0.4987	1 year

Final Model — Confusion Matrix (3year, threshold = 0.1)

	Predicted: Not Bankrupt	Predicted: Bankrupt
Actual: Not Bankrupt	1,200 (TN)	802 (FP)
Actual: Bankrupt	9 (FN)	90 (TP)

Screening Summary

Metric	Value
ROC-AUC	0.8619

Recall	90 out of 99 (90.9%)
Precision	10.1%
F2 Score	0.3494
MCC	0.2180
Companies flagged	892 out of 2,101 (42.5%)
Companies cleared	1,209 out of 2,101 (57.5%)
Bankrupts missed	9 out of 99 (9.1%)

The model reduces analyst workload by **57.5%** (1,209 companies cleared automatically) while catching **90.9%** of all bankruptcies. Only 9 bankrupts are missed.

5. Key Features (SHAP)

Rank	Feature	Financial Meaning
1	3-year GP / Total assets (Attr24)	Long-term profitability
2	(Net profit+Dep)/Liabilities (Attr26)	Debt repayment capacity
3	Quick ratio (Attr46)	Short-term liquidity
4	Quick liquidity * 365 (Attr5)	Days of liquidity coverage
5	Profit on sales / Sales (Attr39)	Operating margin
6	Op. profit / Fin. expenses (Attr27)	Financial leverage coverage
7	Total costs / Sales (Attr58)	Cost efficiency
8	Retained earnings / Assets (Attr6)	Accumulated profitability

SHAP results are economically consistent: profitability, liquidity, and debt coverage are the primary drivers — exactly what banks assess.

6. XGBoost vs Random Forest

Metric	XGBoost	Random Forest	Winner
ROC-AUC	0.8619	0.8158	XGBoost
Recall (threshold 0.1)	0.9091	~0.48	XGBoost
Precision (threshold 0.1)	0.1009	~0.18	Random Forest
F2 Score	0.3494	~0.37	Random Forest (marginal)

MCC	0.2180	~0.24	Random Forest (marginal)
Average Precision (PR-AUC)	0.41	0.21	XGBoost
Peak Lift	~22x	~15x	XGBoost

Corrected comparison: earlier drafts of this table compared XGBoost at its tuned threshold (0.1) against Random Forest at the default threshold (0.5), where RF's recall collapses to 0.00 — an apples-to-oranges comparison. At a matched threshold, Random Forest is competitive on point precision/F2/MCC, but XGBoost is preferred for this use case because it is materially better on the two metrics that matter most for screening: Recall (catching bankruptcies, weighted 2× via the F2 objective) and ROC-AUC / PR-AUC (threshold-independent discrimination, relevant since the deployed threshold can be re-tuned).

XGBoost is selected as the production model: it is the stronger discriminator overall (higher ROC-AUC and PR-AUC) and, at the operating threshold chosen for this screening task, catches substantially more bankruptcies — the priority given the cost of a missed default.

7. Limitations

- **Class imbalance:** Only ~5% bankrupts — leads to low precision.
- **Low precision:** 10.1% at threshold 0.1 — acceptable for screening, not final decisions.
- **Anonymous features:** Attr1-Attr64 without domain labels.
- **Historical data:** 2000-2013 Polish data — may not generalize.
- **No temporal split:** Random train/test — time-based split would be more realistic.